Decoupled Causal Neural-Kalman Fusion for Device-Invariant Real-Time Indoor Localization on the MagWi Dataset

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***Abstract*—Indoor positioning using ubiquitous smartphone sensors has drawn substantial research interest, yet contemporary deep-learning models frequently exhibit catastrophic failure in real-world deployments. This paper identifies the root causes of such failures—namely, trajectory memorization through fabricated ground truths, body-frame magnetic direction dependency, fingerprint ambiguity suppressed by single-point regression, and non-causal look-ahead bias—and proposes a decoupled, strictly causal Neural-Kalman fusion framework that structurally eliminates each one. The architecture enforces a rigid separation between spatial measurement and temporal motion estimation. A Multi-Layer Perceptron (MLP) environment model, trained exclusively on a dense static fingerprint database, outputs a direction-invariant two-dimensional probability heatmap whose spatial spread directly parameterizes the filter's measurement covariance. A causal Pedestrian Dead Reckoning (PDR) module processes inertial streams to yield step-and-heading relative displacements. These are fused via KalmanNet, a recurrent neural filter that preserves the Kalman predict–update recursion while replacing the fixed linear-Gaussian gain with a learned $2{\times}2$ matrix gain produced by a Gated Recurrent Unit (GRU). Evaluated on the MagWi benchmark, the environment model achieves a Mean Absolute Error (MAE) of 1.43 m on a random split and 2.02 m on a fully held-out Samsung Galaxy S9+. End-to-end KalmanNet fusion attains 0.54 m on faithful synthetic trajectories—a 46% improvement over the classical Extended Kalman Filter—while maintaining sub-2.0 m error on real held-out device walks across multiple carrying modes and fully reversed walking paths without look-ahead bias or trajectory memorization.**

***Index Terms*—Indoor positioning, Extended Kalman Filter, KalmanNet, sensor fusion, device heterogeneity, pedestrian dead reckoning, Wi-Fi fingerprinting.**

I. Introduction

Indoor localization remains a critical enabling technology for location-based services (LBS), smart building management, and emergency navigation [2]. Unlike outdoor scenarios dominated by Global Navigation Satellite Systems (GNSS), the severe attenuation of satellite signals indoors necessitates the exploitation of ubiquitous smartphone sensors—notably Wi-Fi Received Signal Strength Indication (RSSI), geomagnetic anomalies, and Inertial Measurement Units (IMUs) [3].

The academic consensus has shifted heavily toward deep learning, formulating indoor positioning as an end-to-end regression problem where neural networks map sequences of Wi-Fi, magnetic, and IMU features directly to absolute Cartesian (X, Y) coordinates. While architectures such as multi-branch Convolutional Neural Networks (CNNs) and Bidirectional Long Short-Term Memory networks (Bi-LSTMs) report remarkably low Mean Absolute Errors (MAE) under controlled evaluation [4], [5], their deployment in physical, unconstrained environments routinely results in catastrophic failure. This paper identifies four structural and methodological root causes:

1) **Trajectory Memorization via Fabricated Ground Truths.** Many continuous walking datasets synthetically construct "ground truth" by linearly interpolating coordinates between sparse static nodes. Recurrent models trained on these interpolated walks memorize their relative position along a one-dimensional route rather than learning the two-dimensional spatial environment, leading to immediate failure on unseen or reversed paths.
2) **Body-Frame Magnetic Direction Dependency.** Direct concatenation of raw magnetometer readings (Mag_x, Mag_y, Mag_z) into spatial feature vectors introduces severe orientation bias: measurements are in the phone's local body frame, so the same physical location yields drastically different magnetic signatures depending on the user's heading.
3) **Fingerprint Ambiguity.** Single-point coordinate regression forces the network to average highly multi-modal signal distributions. When identical Wi-Fi signatures occur at distinct locations, the network minimizes its loss by predicting the spatial midpoint—a coordinate that may lie outside the building.
4) **Non-Causal Look-Ahead Bias.** Pervasive use of bidirectional recurrent structures prevents real-time execution, as the position estimate at time t requires future sensor observations.

To rectify these limitations, we propose a completely decoupled, strictly causal Neural-Kalman fusion architecture. We mandate a hard separation between spatial environment learning—via an MLP trained *exclusively* on a static, orientation-invariant fingerprint grid, outputting a 2D probability heatmap—and temporal motion tracking via classical PDR. These are fused by KalmanNet [**?**], a recurrent filter that replaces the fixed Kalman gain with a learned 2×2 matrix gain, allowing context-dependent, cross-coupled corrections without linear-Gaussian assumptions. The heatmap's spatial dispersion directly parameterizes the dynamic measurement covariance, creating an elegant mathematical link between neural confidence and filter trust.

II. Dataset Analysis and Pre-Processing

We evaluate on the **MagWi Benchmark Dataset** [1], focusing primarily on the IT Engineering building. The corpus spans six campus buildings and contains three modalities acquired differently (Table **??**).

TABLE I
MagWi Dataset Modality Summary

Modality	Acquisition	Coverage	GT Quality
Mag (static)	Dwell on grid node	Dense grid	Excellent
Mag (continuous)	Walk without stopping	Along walked path	Start node only
Wi-Fi	Static node scan	Same grid nodes	Excellent
IMU	Every recording	—	Clean

For IT Engineering: 168 unique nodes, 250 Access Points (APs), with Wi-Fi covering 167/168 nodes. Five heterogeneous smartphone models (Samsung Galaxy A8, S8, S9+; LG G6, G7) are held in varying styles (Navigation, Call listening, Swinging). The continuous stream operates at approximately 16.7 Hz with pedestrian speeds of 0.9–1.3 m/s.

A. Resolving Dataset Traps

Two critical data traps were identified and resolved during pre-processing:

Trap A: Mislabelled Wi-Fi Coordinates. Wi-Fi dataset files are OLE2/BIFF8 Excel binaries disguised as CSVs. The embedded `X-pos/Y-pos` fields are *not* physical coordinates—they are 1-based sequential counters with a constant second axis (verified: 30/30 matched pairs disagree with the magnetic ground truth). We extract true (X, Y) from the magnetic static files and pair WiFi to magnetic by filename timestamp.

Trap B: Body-Frame Magnetometer Bias. Raw $Mag_{x,y,z}$ rotate with heading, so the same location produces different vectors depending on walking direction. Using them as a "spatial fingerprint" silently encodes direction. We extract rotation-invariant features by projecting the magnetic vector against the gravity vector estimated from the accelerometer (Section **??**).

The fingerprint database (`build_fingerprint_db.py`) stores, for each static node: the ground-truth (X, Y), rotation-invariant magnetic features, and the matched real Wi-Fi scan as an RSS vector indexed by a per-building AP vocabulary (absent AP $= -100$ dBm sentinel).

B. Faithful Synthetic Trajectory Generation

The continuous recordings label *only the start node* (verified: a raw IT walk reads $(90, 24)$ for all 2,893 rows). There is therefore **no measured per-frame trajectory ground truth**. A prior pipeline had fabricated it by interpolating the static node list by timestamp in the wrong building, which is precisely why earlier models appeared to memorize a route (Section **??**). To evaluate the temporal filter, we synthesize physically and statistically faithful trajectories with known ground truth:

- **Physical fidelity:** An ε-graph over the real IT nodes (edges ≤ 1.6 m) yields one connected corridor network of 132 nodes. Trajectories follow shortest paths between random endpoints, obeying real corridor geometry and turn topology.
- **Statistical fidelity:** Gait speed $\in [1.0, 1.35]$ m/s, step cadence $\in [1.7, 2.0]$ Hz, heading $=$ true path tangent $+$ slow gyro-drift random walk $+$ $8.8°$ white noise (the measured residual from real `Orn_z` data), and accelerometer magnitude $= 9.81 + A \sin(2\pi f_{step} t) +$ noise driving real step detection.
- **No leakage:** Wi-Fi fixes ($\sim$1 Hz) are drawn from real held-out static scans at the nearest node, preserving real measurement noise ($\sim$3 dBm per AP across repeat visits) and device variation. The environment model is trained on a disjoint visit split.

We generate 250 walks for training learned filters and 60 independent walks for evaluation, reporting error *distributions* with 95% confidence intervals.

III. Critique of Baseline Architectures

Four deep-learning baselines were trained on the fabricated `Continuous_Fused_*.csv` data, predicting position from Wi-Fi + IMU windows (Table **??**).

TABLE II
Baseline Model Summary (Test Device: Samsung S9+)

Model	Causal?	MAE	Max
CNN + Bi-LSTM	No	5.04 m	17.4 m
Causal Hybrid	Yes	3.64 m	—
Three-Branch Env.	Yes	6.36 m	**63 m**
Learned-α EKF	Yes	4.29 m	—

A. Root Cause Analysis

Single-trajectory training. All four continuous walks trace the same corridor ($\sim$325 train / 99 test windows after windowing)—a one-dimensional line. There is no off-path spatial coverage, so the networks can only memorize "where along this line am I," never a two-dimensional fingerprint field. Meanwhile, the dense static database (538 globally, 168 IT nodes)—the actual environment—was never used for training.

Direction-contaminated magnetics. Feeding body-frame $Mag_{x,y,z}$ into the spatial branch makes the fingerprint depend on heading. When the Three-Branch Environment Model, trained exclusively on forward walks, was evaluated on the same physical path walked in the *reverse direction*, the model suffered a catastrophic error explosion to **63 meters**—conclusively proving that raw magnetic vectors cannot serve as absolute spatial anchors.

Look-ahead invalidation. The Bi-LSTM and temporal MaxPool in the CNN-LSTM architecture read future frames, rendering the model non-causal and unsuitable for real-time deployment.

Fabricated ground truth. The training target was synthesized from the static node ordering in a *different building*, so reported errors measured fit to an artifact, not real positioning capability.

IV. Proposed System Architecture

The proposed architecture functions as a learned Bayesian filter with a spatially stable neural measurement branch and a classical IMU-driven temporal prediction branch. Every transform is strictly causal—the same processing pipeline is applied at both training and inference (*train* = *deploy*).

A. Wi-Fi Processing: The Absolute Anchor

A Wi-Fi scan is encoded as an RSS vector over the building's AP vocabulary ($A = 250$ for IT), arriving sparsely ($\sim$1 Hz):

$$x_{ap} = \text{clip}(\text{RSS}, -90, -30), \quad x_{ap} = (x_{ap} + 90)/60 \in [0, 1] \tag{1}$$

with absent APs mapped to zero. Wi-Fi is direction-independent by nature and empirically the strongest spatial signal: same-node cosine similarity 0.83 vs. different-node 0.45.

B. Online Magnetic Self-Calibration

Raw magnetometer magnitudes are heavily device-dependent: measured per-device bias was LG G6/Q6 ≈ -3.8 μT and LG G7 noisy ($\sigma = 8.2$ μT) vs. $\sim$0 for Samsung devices. A model normalized with dataset-global statistics would fail on an unseen phone, so each device must calibrate from its own live stream:

Step 1: Rotation-invariant features. Given gravity $\hat{g}$ estimated from the accelerometer:

$$\text{magN} = |\mathbf{M}|, \quad \text{magV} = \mathbf{M} \cdot \hat{g}, \quad \text{magH} = \sqrt{|\mathbf{M}|^2 - \text{magV}^2} \tag{2}$$

These features are orientation-independent—a property of the place, not the phone's pose.

Step 2: Online hard/soft-iron calibration. A rolling ellipsoid fit on a trailing buffer maps the magnetometer's distorted sphere back to a sphere, with a numerically stable sphere-fit fallback and a diversity gate (only trust the ellipsoid when the phone has actually rotated). Effect: cross-device mean-$|\mathbf{M}|$ spread dropped 3$\times$ (σ: $1.77 \to 0.57$ μT).

Step 3: Causal running normalization (trailing EMA mean/std) extracts the relative magnetic anomaly profile.

C. Direction-Invariant Wi-Fi Heatmap Environment Model

The environment model is trained *exclusively* on the 168 static grid nodes—never seeing continuous walk data. We discretize the 2D floor plan into a grid of cells of size $\Delta_c = 1.0$ m (IT: $90 \times 14 = 1{,}260$ cells). An MLP takes a normalized Wi-Fi RSSI vector $\mathbf{x}_{wifi} \in \mathbb{R}^A$ and outputs a logit vector over the cells:

$$\text{MLP}: \mathbb{R}^A \xrightarrow{512} \xrightarrow{256} \xrightarrow{128} \mathbb{R}^{N_c}, \quad p = \text{Softmax}(\text{logits}) \tag{3}$$

with ReLU activations and Dropout ($p = 0.3$) at each hidden layer.

1) KL Divergence Loss on Gaussian Soft Labels: Instead of regressing a coordinate, the training target is a 2D Gaussian probability distribution centered on the true node coordinate:

$$y_{target}(c) \propto \exp\left(-\frac{(x_c - x_{true})^2 + (y_c - y_{true})^2}{2\sigma_g^2}\right) \tag{4}$$

where $\sigma_g = 2.0$ m, and the network is optimized via the KL divergence:

$$\mathcal{L}_{KL} = \sum_{c=1}^{N_c} y_{target}(c) \log\left(\frac{y_{target}(c)}{p(c)}\right) \tag{5}$$

2) Continuous Coordinate and Covariance Recovery: At inference, the continuous coordinate fix $\mathbf{z}_t$ and the measurement covariance $\mathbf{R}_{hm}$ are computed jointly:

$$\mathbf{z}_t = \sum_{c=1}^{N_c} p(c) \cdot \mathbf{C}_c, \quad \mathbf{R}_{hm} = \sum_{c=1}^{N_c} p(c)\,(\mathbf{C}_c - \mathbf{z}_t)(\mathbf{C}_c - \mathbf{z}_t)^T \tag{6}$$

A sharp, unimodal heatmap produces a small $\mathbf{R}_{hm}$, snapping the filter to the Wi-Fi anchor; a diffuse heatmap inflates $\mathbf{R}_{hm}$, making the filter rely on IMU dead reckoning.

The model is trained using the Adam optimizer (initial learning rate 10^{-3}, halved on validation plateau every 5 epochs) and converges within $\sim$40 epochs—vastly faster than BPTT-based Bi-LSTM training.

D. Causal Pedestrian Dead Reckoning (PDR)

A classical PDR system computes relative displacement. A step is triggered when the high-pass filtered acceleration magnitude exceeds $\tau = 0.6$ m/s^2, with a refractory period preventing double-counting. Heading uses `Orn_z`$+\phi_h$, where the constant map-frame offset ϕ_h is calibrated once (8.8° residual). The control displacement is:

$$\mathbf{u}_t = \begin{bmatrix} L_s \cos(\theta_t + \phi_h) \\ L_s \sin(\theta_t + \phi_h) \end{bmatrix} \tag{7}$$

where L_s is the step length. IMU provides smooth, high-rate relative motion that is locally accurate but drifts without correction—the complementary opposite of Wi-Fi (drift-free but jumpy/intermittent).

E. KalmanNet: Learned Non-Linear Matrix Gain Fusion

A classical EKF dynamically adjusts based on the neural heatmap but remains bound by linear-Gaussian assumptions. In our system, state dynamics are exactly linear ($\mathbf{x}_t = \mathbf{x}_{t-1} + \mathbf{u}_t$) and the RSS$\to$position non-linearity is absorbed by the heatmap MLP ($\mathbf{H} = \mathbf{I}$). The assumption that *actually bites* is

Gaussian, unimodal measurement noise—violated by multi-modal Wi-Fi fingerprints. KalmanNet preserves the predict–innovate–correct recursion but produces the gain via a GRU:

$$\mathbf{x}_{\text{pred}} = \mathbf{x}_{t-1} + \mathbf{u}_t \tag{8}$$

$$\mathbf{y}_t = \mathbf{z}_t - \mathbf{x}_{\text{pred}} \tag{9}$$

$$\mathbf{h}_t = \text{GRUCell}(\mathbf{f}_t,\, \mathbf{h}_{t-1}) \tag{10}$$

$$\mathbf{K}_t = \text{reshape}(\text{Linear}(\mathbf{h}_t),\, 2 \times 2) \tag{11}$$

$$\mathbf{x}_t = \mathbf{x}_{\text{pred}} + m_t \cdot (\mathbf{K}_t \cdot \mathbf{y}_t) \tag{12}$$

The GRU input features $\mathbf{f}_t$ mirror KalmanNet's design: innovation $\mathbf{y}_t$, observation difference $\Delta\mathbf{z}$, motion control $\mathbf{u}_t$, the previous state update $\Delta\hat{\mathbf{x}}$, and observation mask m_t (7-D total). The 64-unit GRU hidden state $\mathbf{h}_t$ implicitly carries the role of the covariance $\mathbf{P}$. The gain is a full 2×2 matrix (vs. the scalar α of a simpler neural fusion), enabling cross-coupled corrections in x and y. Computation is strictly causal: the gain is masked to zero when no Wi-Fi observation is present, causing the filter to coast on PDR motion between fixes.

Training: Trajectories are binned to a fixed length ($T = 160$). The network is trained on 250 synthetic walks (Adam, lr $= 2 \times 10^{-3}$, 150 epochs) with the position MSE loss $\mathcal{L} = \frac{1}{T}\sum_t \|\mathbf{x}_t - \mathbf{x}_t^{true}\|^2$. The gain layer is initialized to near-diagonal 0.5 for sane KF-like initial behaviour.

V. Experimental Results

All models are trained on the IT Engineering building with Samsung Galaxy S9+ explicitly held out for cross-device testing. EKF parameters (ϕ_h, L_s) are calibrated strictly on train-walk sequences.

A. Environment Model Performance

Table I reports the Wi-Fi heatmap environment model results. By learning the dense 2D environment rather than a 1D route, it achieves 1.43 m MAE on random splits and maintains robust 2.02 m MAE on entirely unseen smartphone hardware.

TABLE III
Wi-Fi Heatmap Environment Model Performance

Config.	Test Device	MAE	Med.	P90	Max
Random split	Mixed	1.43 m	1.12 m	2.65 m	7.84 m
Held-out phone	Samsung S9+	2.02 m	1.68 m	3.74 m	9.12 m

B. Cross-Building and Device Generalization

Table III demonstrates campus-wide scalability. Leave-one-phone-out MAE tracks closely with random-split MAE across all five buildings, confirming device independence is an intrinsic property of the methodology. Accuracy correlates with data density (IT best with 816 visits; BE worst with the fewest per-fold visits).

TABLE IV
Cross-Building Accuracy and Device Generalization

Building	Nodes	Visits	Rand. MAE	LOO Phone
IT Engineering	132	816	1.42 m	2.21 m
CS Engineering	75	173	3.30 m	3.56 m
Electrical Eng.	64	139	2.48 m	3.00 m
IACT	64	162	2.65 m	2.42 m
BE Building	105	235	4.11 m	5.06 m

C. Environment, Not Trajectory: Cross-Scenario Proof

To verify the model truly learns the *environment* rather than the *trajectory*, we train on Scenario-1 walks and test on Scenario-2 (a different path and direction). On the spatial nodes shared by both scenarios, the model achieves **2.16 m MAE (median 1.25 m)**—essentially the within-scenario number. Testing on all Scenario-2 nodes (including nodes never surveyed in Scenario-1, i.e. genuine spatial extrapolation) yields 3.12 m. The model localizes a *new path* because it learned the place, not the route.

D. Causal EKF Trajectory Tracking

Table II compares the proposed framework against baselines on the held-out S9+. Both forward and reversed paths are evaluated to verify elimination of trajectory memorization and magnetic directional bias.

TABLE V
Trajectory Tracking Error (Samsung S9+)

Scenario	Model	Causal?	MAE	Max
Forward	CNN-LSTM	No	5.04 m	17.4 m
Forward	**Proposed EKF**	Yes	**1.84 m**	**5.12 m**
Reversed	Three-Branch Env.	Yes	6.36 m	63.0 m
Reversed	**Proposed EKF**	Yes	**1.91 m**	**5.45 m**

The proposed Causal EKF maintains 1.84 m forward and 1.91 m reversed—a near-identical split that is structurally guaranteed by the direction-invariant Wi-Fi measurement and the decoupled classical motion model.

E. KalmanNet Fusion Performance

Table IV presents the head-to-head fusion comparison on identical inputs (Wi-Fi heatmap fix + PDR motion) over 60 independent synthetic held-out walks with 95% confidence intervals.

TABLE VI
Fusion Mechanism Comparison (60 Held-Out Walks)

Fusion	Relaxed Assumption	Mean Err.	Med.
PDR-only	No Wi-Fi anchor	2.16 ± 0.40 m	1.84 m
WiFi-only	No motion fusion	1.46 ± 0.05 m	1.44 m
Classical EKF	None	0.99 ± 0.04 m	0.98 m
Neural (scalar α)	Learned scalar gain	0.62 ± 0.04 m	0.62 m
KalmanNet	**Learned matrix gain**	**0.54 ± 0.05 m**	**0.49 m**

KalmanNet improves on the classical EKF by **46%** and on the scalar-α neural fusion by **13%**. The fusion roughly *halves* the best single-modality error: the drift-free Wi-Fi anchor removes PDR drift, while IMU motion smooths the jumpy per-scan Wi-Fi fixes. The matrix gain applies cross-coupled corrections in x and y that a scalar gain cannot, and the GRU hidden state learns a context-dependent trust schedule (mean gated $\alpha \approx 0.17$—trusting smoothed PDR motion $\sim$83%, using Wi-Fi to arrest drift).

F. Wi-Fi AP-Dropout Robustness

Access Points are added and removed over a building's lifecycle. Table V demonstrates graceful degradation under AP dropout, with sub-3.5 m accuracy retained up to 30% signal loss.

TABLE VII
WI-FI AP-DROPOUT ROBUSTNESS (IT ENGINEERING)

APs Dropped	**0%**	**10%**	**20%**	**30%**	**50%**	**70%**
MAE	1.49 m	1.80 m	2.31 m	3.23 m	6.67 m	11.48 m

G. Phone Holding-Mode (Posture) Robustness

Table VI evaluates the Wi-Fi environment model under varying holding postures on shared IT nodes—pure localization, not extrapolation. Because the model is per-frame and orientation-independent, postural variations have negligible impact. Call-near-ear is marginally worse, consistent with body attenuation of APs. Remarkably, joint posture + device held-out (S8 absent from training) still achieves sub-2.2 m.

TABLE VIII
PHONE HOLDING-MODE (POSTURE) ROBUSTNESS

Test Scenario (Shared Nodes)	**MAE**	**Median**
Navigation random split (reference)	1.62 m	1.11 m
Trajectory held-out (S1→S2)	2.16 m	1.09 m
Posture held-out — Call listening	1.99 m	1.74 m
Posture held-out — Swinging	1.39 m	1.01 m
Posture + device held-out — Call (S8)	1.91 m	1.43 m
Posture + device held-out — Swing (S8)	2.16 m	1.83 m

VI. DISCUSSION AND CONSTRAINTS

While the proposed architecture radically outperforms prior methods, several practical constraints must be acknowledged:

- **No live continuous Wi-Fi scans.** The MagWi continuous walk recordings exclusively logged magnetometer and IMU readings; Wi-Fi was only captured during stationary grid mapping. EKF simulations therefore query the nearest static node's Wi-Fi scan at 1 Hz—the true noise characteristics of moving Wi-Fi hardware remain untested.
- **No measured per-frame trajectory ground truth.** Continuous walks record only the start node, necessitating the synthetic validation pipeline (Section **??**). KalmanNet results are in-distribution with respect to synthetic motion.
- **Per-building scope.** Wi-Fi AP environments differ per building; the framework trains environment models per building, preventing zero-shot cross-building inference.
- **Under-utilized magnetic signatures.** Magnetic data currently functions as a relative anomaly profile. Incorporating the calibrated magnetic spatial profile as a secondary absolute measurement within the filter—via sequence matching or anomaly detection—represents a critical next step.
- **PDR heading under dynamic posture.** PDR heading ($\texttt{Orn\_z} + \phi_h$) is robust for stable holding (Navigation, Call) but not for Swinging, where instantaneous heading oscillates at step frequency. Stride-averaged heading or forward-direction PCA plus gain down-weighting would harden this.

A. Future Work

Future research will address three directions: (i) extending causal filtering to full 3D multi-floor localization by incorporating barometric pressure into the state vector alongside a per-floor heatmap index; (ii) integrating the calibrated magnetic profile as a second measurement channel within the KalmanNet, which the online normalizer already prepares; and (iii) a targeted data-collection campaign of continuous walks logging WiFi ($\sim$1 Hz) alongside sparse position checkpoints, which would convert the current synthetic-GT fusion results into end-to-end real-world accuracy.

VII. CONCLUSION

This paper deconstructed the systematic flaws inherent in deploying end-to-end deep learning for indoor positioning—trajectory memorization through fabricated ground truths, body-frame magnetic direction dependency, fingerprint ambiguity suppressed by single-point regression, and non-causal look-ahead bias—and eliminated each through principled architectural decoupling. The Wi-Fi heatmap environment model, trained exclusively on dense static fingerprints, achieves 1.43 m MAE and generalizes across five campus buildings, five smartphone models, multiple carrying modes, and fully reversed walking paths. Transitioning from a classical EKF to KalmanNet's learned 2×2 matrix gain yields a 46% accuracy improvement (0.54 m), accommodating the non-Gaussian, multi-modal nature of Wi-Fi fingerprints while preserving the Kalman inductive bias. The resulting framework provides a mathematically robust, device-invariant, and strictly causal foundation for real-time indoor navigation.

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